Course Objectives:

Understand database concepts, relational models, SQL, normalization, transactions, and database design.

Course Content:

UNIT 1: Introduction to Databases

- Database Concepts
- DBMS Architecture
- Data Models
- Database Users
- Schema and Instances

UNIT 2: Relational Model and SQL

- Relational Algebra
- SQL Basics
- Joins and Subqueries
- Views
- Constraints

UNIT 3: Database Design

- ER Model
- ER Diagrams
- Mapping ER to Relational Model
- Functional Dependencies
- Normalization

UNIT 4: Transaction Management

- Transactions
- ACID Properties
- Concurrency Control
- Deadlocks
- Recovery Techniques

UNIT 5: Advanced Database Concepts

- Indexing
- Query Optimization
- Distributed Databases
- NoSQL Databases
- Data Warehousing

Suggested Books:

- Korth — Database System Concepts
- Elmasri & Navathe — Fundamentals of Database Systems

Course Outcomes:

- Design relational databases.
- Write optimized SQL queries.
- Apply normalization techniques.
- Understand transaction management.
